**PROJECT DOCUMENTATION: END-TO-END WEATHER ANALYTICS SYSTEM**

**Domain:** Data Engineering & Business Intelligence

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## 1. Executive Summary

The goal of this project is to build an automated system that extracts real-time weather data, processes it through an ETL (Extract, Transform, Load) pipeline, and visualizes historical trends. The system is designed to be fully autonomous, requiring zero manual intervention for data collection and reporting.

## 2. Problem Statement

Monitoring weather patterns manually is inefficient, and analyzing long-term seasonality (e.g., the last 12 months) is impossible without a structured historical database. This project solves these challenges by:

1. Automating live data collection from global sources.
2. Building a relational database to store time-series data.
3. Providing interactive insights into correlations such as Humidity vs. Rainfall and Seasonal Extremes.

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## 3. System Architecture

The architecture is divided into four distinct layers:

### A. Data Acquisition Layer

- **Source:** OpenWeatherMap API.
- **Scope:** 15+ Major Indian Cities (Metros and Tier-1 cities).
- **Ingestion Frequency:** Scheduled 1-Hour refresh intervals.

### B. ETL Layer (Python)

- **Extraction:** Utilized the requests library to fetch JSON payloads from REST API endpoints.
- **Transformation:** Data cleaning performed using Pandas. This included Kelvin to Celsius conversion, timestamp standardization (ISO format), and data validation.

- **Loading:** Data is ingested into an RDBMS (SQLite/MySQL) using the sqlite3 and SQLAlchemy engines.

**C. Storage Layer (Database)**

- **Architecture:** Star Schema design for optimized analytical querying.
- **Fact Table:** weather_data (Stores metrics like Temp, Humidity, Pressure, Wind Speed).
- **Dimension Tables:** cities (Geospatial data) and Calendar_Table (Time Intelligence).

**D. Visualization Layer (Power BI)**

- **Connectivity:** Integrated via Python Script Connector.
- **Modeling:** Established One-to-Many relationships between dimensions and fact tables.
- **Analytics:** Leveraged DAX (Data Analysis Expressions) for complex business logic.

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**4. Technical Specifications**

**Database Schema**

| Table Name | Attributes | Description |
|---|---|---|
| **cities** | city_id, city_name, lat, lon | Master location data |
| **weather_data** | city_id, temp, humidity, pressure, timestamp | Fact records of weather metrics |
| **Calendar_Table** | Date, Year, Month, Season, SortKey | Time dimension for advanced filtering |

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**5. Advanced Analytics & DAX Measures**

Several advanced calculations were implemented to drive the dashboard:

- **Seasonal Categorization:** Logical grouping of months into Summer, Monsoon, and Winter.
- **Chronological Sorting:** Created a Month_Sort_Key to ensure the dashboard prioritizes the most recent month.
- **Latest Condition Tracking:** Developed a measure using MAX(timestamp) to display the most recent weather status on a KPI card.

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**6. Dashboard Features & Insights**

1. **Seasonal Heatmaps:** Visualized temperature variations across different cities and months.
2. **Correlation Analysis:** Investigated the relationship between rising humidity and rainfall frequency.
3. **Trend Forecasting:** 12-month historical line charts to identify annual weather peaks and troughs.
4. **Extreme Weather Alerts:** Automated identification of the highest and lowest recorded temperatures per season.

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**7. Challenges & Resolutions**

- **Challenge:** Encountered Circular Dependency issues while creating time-based calculated columns.
- **Resolution:** Implemented a dedicated "Calendar Table" and utilized "Sort by Column" features to decouple the logic.
- **Challenge:** Historical data constraints within the Free API Tier.
- **Resolution:** Developed a Python-based data simulator to generate 365 days of realistic, seasonal mock data for comprehensive stress testing.

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**8. Conclusion**

This project demonstrates a scalable data pipeline capable of handling real-world time-series data. The current infrastructure is cloud-ready and serves as a foundation for future Predictive Analytics using Machine Learning (e.g., Rainfall Prediction).